SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B2. SYMMETRY AND RELATIVITY

MICHAELMAS TERM 2020

Tuesday, 06 October

Opening time: 09.30 am (BST)

You have two hours and thirty minutes to complete the paper and upload your answer file

Answer **two** *questions.*

Start the answer to each question on a fresh page.

A list of physical constants and conversion factors is available at the Physics website.

You are permitted to use calculators.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Page 2 contains physics formulae and data for this paper.
The questions start on page 3.

The following notation is used throughout the paper: capital bold letters (e.g., $\mathbf{U}$) indicate 4-vectors; small case bold letters (e.g., $\mathbf{v}$) indicate 3-vectors.

However the symbols $\mathbf{E}$ $\mathbf{B}$ and $\mathbf{A}$ exceptionally indicate 3-vectors, being the electric and magnetic fields and the magnetic vector potential, respectively.

The Einstein summation convention, the Minkowski metric, and other conventions adopted and used in the course are applied.

1. (a) A particle of mass m collides with a free stationary particle of mass M. After the collision, we are left with N particles with masses m_i ($i = 1, \ldots, N$). Derive an expression for the minimum total energy E of the incident particle required to create the collision products, in terms of m, M and $\sum_i m_i$. Throughout the rest of this question, we will assume that this minimum energy condition is just met. [6]

(b) An example of such a collision is where two protons collide to produce pions via the process: $p + p \to p + p + \pi^0$. Using your earlier result, estimate the minimum kinetic energy of the incident proton required, and compare it to the pion's rest mass energy. Compute the velocity v of the pion in the lab frame. [6]

(c) After a short time, the neutral pion decays into two photons. Compute the wavelength of the photons in the pion's rest frame. Over a large number of such collisions, the photons are emitted isotropically in the pion's rest frame. Show that, in the laboratory frame, half the photons are emitted into a cone with half-angle $\cos^{-1}(v/c)$. [8]

(d) How can the kinetic energy of the colliding protons required to create a pion be further reduced, and by how much? Describe the state of the particles after the collision. [5]

2. (a) Define the 4-velocity $\mathbf{U}$ and the 4-force $\mathbf{F}$. Show that a body subject to a pure force obeys $\mathrm{d}E/\mathrm{d}t = \mathbf{v} \cdot \mathbf{f}$, where E and $\mathbf{v}$ are the energy and the 3-velocity of the body, respectively, and $\mathbf{f}$ is the 3-force being applied to it. [6]

(b) A rocket accelerates at a constant rate a_0 in its instantaneous rest frame. Write down an equation of motion for the rocket in the lab frame, in terms of its rapidity η, where $\cosh\eta = \gamma$. [Hint: you may use, without proof or derivation, the result $A^\mu A_\mu = c^2 \dot{\eta}^2 \cosh^2 \eta$, where A^μ is the 4-acceleration.] Show that, for a suitable choice of origin, the trajectory of the rocket in the lab frame can be written:

$$x^2 - c^2t^2 = (c^2/a_0)^2.$$

Using a diagram or otherwise, show that a photon emitted at ($t = 0$, $x = 0$) in the $+x$ direction never catches up with the rocket. Comment on this result. [10]

(c) If the rocket propels itself by emitting photons (a 'photon rocket'), and is initially at rest with mass m, show that by the time it reaches speed v its mass has dropped to αm, where

$$\alpha = \sqrt{\frac{1-\beta}{1+\beta}},$$

where $\beta = v/c$. If the rocket is to accelerate to a Lorentz factor of 8 and then decelerate to rest at its destination, what fraction of its initial mass must be devoted to fuel? Without detailed calculation, explain why this fraction is even larger if the rocket emits massive particles rather than photons. [9]

3. (a) Write down the components of the Maxwell field tensor $F^{\mu\nu}$ in terms of the components of the electric field **E** and magnetic field **B**. Show that two of Maxwell's equations in 3-vector form can be written in terms of the field tensor as:

$$\partial_\nu F^{\mu\nu} = \mu_0 J^\mu$$

where μ_0 is the magnetic permeability of free space and J^μ is the 4-current. Use the field tensor to obtain an invariant relationship between the magnitudes of **E** and **B**. [8]

(b) The field tensor can also be written

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu,$$

where $A^\mu(\phi/c, \mathbf{A})$ is the 4-potential, and

$$\begin{aligned} \mathbf{B} &= \nabla \times \mathbf{A}, \\ \mathbf{E} &= -\nabla\phi - \tfrac{\partial \mathbf{A}}{\partial t}. \end{aligned}$$

Show that these definitions ensure that the fields are gauge invariant and that Maxwell's equations reduce to a single tensor equation:

$$\partial_\nu \partial^\nu A^\mu = -\mu_0 J^\mu,$$

for a suitable choice of gauge. [9]

(c) A straight wire with linear charge density ρ in its rest frame moves with velocity **v** along the length of the wire. Find the electric and magnetic fields generated by the wire in the rest frame S of the wire, and in the (stationary) laboratory frame S'. [You may state the Lorentz transformation equations for the fields without proof.] Is there a frame in which the electric field is zero? [8]

4. (a) Write down an expression for the 4-potential A^μ of an electromagnetic wave travelling in vacuum, defining all the quantities involved. Use this to show that the phase ϕ of the wave is Lorentz invariant. Show that a single, isolated electron propagating in a vacuum cannot emit a photon. [8]

(b) Two events in laboratory frame S are characterised by 4-coordinates $X_1 = (ct_1, \mathbf{x_1})$ and $X_2 = (ct_2, \mathbf{x_2})$. Write down the condition for these events to be connected by a space-like interval. Can we find an inertial frame S' where these events are occuring simultaneously? [4]

(c) Two photons of the same angular frequency ω and with 4-momenta $\mathbf{P_1}$ and $\mathbf{P_2}$ move in the lab frame in the $+x$ and $+y$ direction, respectively. Find the rest mass of the system as a function of ω and the velocity of the centre of mass frame relative to the lab frame. [5]

(d) Show that if two particles have velocities **u**, **v** relative to some frame, then the speed of one particle relative to the other is:

$$w = \frac{\sqrt{(\mathbf{u}-\mathbf{v})^2 - (\mathbf{u}\times\mathbf{v})^2/c^2}}{1 - \mathbf{u}\cdot\mathbf{v}/c^2}.$$

[Hint: you may find it useful to show that, for any vectors **a**, **b**, $(\mathbf{a}\times\mathbf{b})^2 = a^2b^2 - (\mathbf{a}\cdot\mathbf{b})^2$.] [8]